

Software Requirements System Document

**Project Title: Blast Assist, Safety
Blasting Arrangement App**

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1. Introduction

An app that helps blasting workers calculate blasting materials fast for a specific mine. It also targets community members by informing them that the next blasting is going to occur for them to stay on the safe line.

1.1 Purpose

The purpose of this document is to provide a definitive and comprehensive description of the **Blast Master Pro** mobile application. This Software Requirements Specification (SRS) serves as the primary technical and operational roadmap for the project, ensuring that all engineering, safety, and community-related goals are aligned before development begins. Specifically, this document aims to:

For the Development Team: *Define the precise calculation logic for predictive blasting variables (Depth, Spacing, and Explosive Type) and the technical architecture for the **Firestore-backed** offline synchronization and **GIS mapping** integration.*

For Project Stakeholders: *Outline how the application will improve operational efficiency by automating manual calculations, reducing human error, and providing a historical database of blasting activities to optimize future mining costs.*

For Mining Safety Inspectors: *Detail the high-priority "Force Notification" system and GIS-based safety radius monitoring. This ensures that the app meets local safety regulations by providing a reliable, verifiable method for protecting both site workers and the surrounding community from blasting hazards.*

Ultimately, this SRS acts as a formal agreement to ensure that **Blast Master Pro** transforms the traditional blasting process into a data-driven, safety-first digital workflow.

1.2 Scope

Blast Master Pro is a specialized digital ecosystem designed to bridge the gap between complex mining engineering and community safety. The application's reach is defined by the following three core modules:

Predictive Blasting Engine (Operational Efficiency)

The primary function of the application is to serve as a high precision calculation tool to optimize blasting parameters.

- **Input-Driven Prediction:** Users can input specific known variables such as the total **area** to be blasted, the **landscape type** (e.g., rocky or mountainous), and the **explosive type**.

- **Automated Output:** The engine processes these variables to determine critical technical requirements, specifically **hole depth**, **hole spacing**, and the precise **quantity of explosives** required.
- **Smart Logic:** The system is designed to work with a "minimum variable" logic if a user provides at least two variables, the app calculates and predicts the remaining unknown parameters.

Digital Data Management (*Continuous Improvement*)

The app replaces manual logging with a digital historical database powered by **Firestore Storage**.

- **Historical Archiving:** Every blast performed is recorded with all associated variables, including depth, distance, landscape, and explosive specifications.
- **Performance Review:** Engineers can review past blasting data to analyse patterns and improve the efficiency of future operations.
- **GIS Documentation:** Every record is stamped with **GPS coordinates**, capturing the exact ground data at the time of the blast for audit purposes.

Automated Safety Integration (*Community Protection*)

Beyond engineering, the app serves as a critical safety link between the mine and the public.

- **Forced Notification System:** This is an automated, high-priority alert system triggered before a blast occurs.
- **Perimeter Awareness:** The system identifies registered residents and individuals living near the mine perimeter or those heading toward the mine on a scheduled blasting day.
- **Public Information Hub:** Provides a limited-access interface for the community to view updates on upcoming blasts and watch videos of past activities, ensuring transparency and safety compliance.

Remote Site Management

- **GIS-Based Mapping:** Engineers can locate and mark blasting areas on a map without being physically present on-site.
- **Spatial Visualization:** Using the **Google Maps API**, the system allows for the visualization of safety zones and nearby infrastructure to ensure accurate planning.

1.3 Definitions, Acronyms, and Abbreviations.

Technical Mining Terms

- **ANFO:** Ammonium Nitrate/Fuel Oil; a widely used bulk industrial explosive.
- **Blasting Pattern:** The spatial arrangement of drill holes, including burden and spacing.
- **Burden:** The distance between the drill hole and the nearest free face of the rock.
- **Fragmentation:** The degree to which rock is broken into smaller pieces by a blast.
- **Landscape Profile:** The topographical characteristics of the blast site (e.g., Mountainous vs. Rocky).
- **Rock Density:** The mass per unit volume of the rock being blasted, used to calibrate explosive energy.
- **Spacing:** The distance between adjacent drill holes in a row.
- **Stemming:** Inert material (like gravel or oil) placed in the top of a drill hole to confine explosive gases.

Software & Technology Terms

- **AES (Advanced Encryption Standard):** A symmetric-key block cipher used to protect sensitive data.
- **API (Application Programming Interface):** A set of protocols for building and integrating application software.
- **FCM (Firebase Cloud Messaging):** A cross-platform messaging solution that lets you reliably send messages at no cost.
- **Firestore:** A flexible, scalable NoSQL cloud database used to store and sync data.
- **GIS (Geographic Information System):** A system designed to capture, store, and analyze geographic and spatial data.
- **GNSS (Global Navigation Satellite System):** The standard generic term for satellite navigation systems (e.g., GPS).
- **GPS (Global Positioning System):** A satellite-based radionavigation system.
- **HIG (Human Interface Guidelines):** Software development documents which offer developers a set of recommendations for UI/UX (specifically for iOS).
- **JWT (JSON Web Token):** A compact, URL-safe means of representing claims to be transferred between two parties.

- **Local-First:** A design pattern where the application prioritizes local data storage over cloud storage to ensure offline functionality.
- **NoSQL:** A non-relational database used for storing large volumes of data that doesn't follow a fixed schema.
- **RBAC (Role-Based Access Control):** A method of restricting system access to authorized users based on their role (e.g., Engineer vs. Community Member).
- **SDK (Software Development Kit):** A collection of software development tools in one installable package.
- **SRS:** Software Requirements Specification.

1.4 Domain and Problem

Domain: Mining Engineering

Problem: Blasting workers currently estimate materials manually, leading to inefficiency and safety risks. Community members near mines lack real-time blast alerts. This app automates calculations and provides location-based safety notifications.

2. The overall Description

2.1 Product Perspective

Interfaces

The system adopts a **dual-persona architecture** to ensure data integrity and user-specific experiences:

- **Engineer UI:** A dashboard centric interface focused on data entry, mathematical precision, and map manipulation. It includes secure authenticated states to prevent unauthorized blasting triggers.
- **Community UI:** A simplified, "read-only" gateway. It prioritizes high visibility alerts (Force Notifications) and a media gallery for public transparency.

Hardware Interfaces

- **GPS Receiver:** The app must interface with the mobile device's GNSS (Global Navigation Satellite System) to provide sub-5-meter accuracy for blast site tagging.
- **Notification Engine:** Utilizes the device's vibration motor and high-priority audio channels to ensure "Force Notifications" are noticed even if the phone is in a pocket or near loud machinery.

Software Interfaces

- **Google Maps SDK:** Provides the base layer for satellite imagery and coordinate conversion.
- **Firebase Authentication:** Handles secure JWT (JSON Web Token) sessions for engineers.
- **Firebase Firestore:** A NoSQL real-time database used to store blasting logs and community contact lists.
- **Firebase Cloud Messaging (FCM):** The engine used to push instantaneous alerts to community members.
- **Memory Constraints**
- **Local Caching:** Since it works in remote mines, the app must reserve a maximum of **500MB** of local storage for "Offline Map Tiles" and a local **SQLite** database to store calculation logs until the next sync.
- **RAM Management:** The calculation engine must be optimized to run on devices with as little as 2GB of RAM without crashing other background safety processes.

Operations

- **Synchronization Logic:** The app operates on a "Local-First" principle. In a mine (zero-signal), data is committed to the local device. The **WorkManager API** (for Android) or **Background Tasks** (for iOS) will automatically push these updates to the cloud once a 3G/4G or Wi-Fi signal is detected.

Site Adaptation Requirements

- **Environmental Profiles:** The calculation engine is not "one-size-fits-all." It includes a configuration layer where engineers can select **Rock Density Profiles** (e.g., Soft Sandstone vs. Hard Granite) and **Landscape Slopes** (Mountainous vs. Flat) to calibrate the spacing and depth formulas.

2.2 Product Assumptions

- **Device Readiness:** It is assumed community members have "Push Notifications" enabled for this app as a condition of use for their safety.
- **Engineer Accuracy:** It is assumed the user understands explosive safety protocols; the app provides calculations, but the final decision remains with the certified engineer.

2.3 User Characteristics

- **Mining Engineers/Workers:** These users are often wearing gloves or working in bright sunlight; therefore, the interface must have high-contrast buttons .
- **Community Members:** Likely have varying levels of digital literacy. The interface must use clear icons (e.g., a "Red Alert" icon) rather than complex text descriptions.
- **Mining Engineers:** High-level access; authorized to use the Predictive Calculator, GIS Mapping, and History.

- **Mine Workers:** Mid-level access; authorized only for GIS Mapping (site location) and History (record review). No access to calculation/trigger functions.

2.4 Constraints

- **Zero-Signal Tolerance:** The core calculator and GPS logging must have **zero dependencies** on a live server to prevent site delays.
- **Regulatory Compliance:** All data storage must comply with regional **Data Privacy Acts** (like GDPR or POPIA) regarding the storage of community phone numbers and location data.

2.5 Assumption Dependencies

The success of the "Force Notification" system is dependent on **Local Telecommunication Infrastructure**. If a local cell tower is down, the app's ability to warn community members is limited. The system assumes that the mine's external communication office has a stable connection to trigger the cloud-based notification broadcast.

2.6 Apportioning of Requirements

In the event of a delayed development timeline, requirements will be prioritized as follows:

- **Phase 1 (Critical):** Offline Calculator, Local GPS Logging, and Basic Firebase Auth.
- **Phase 2 (Safety):** Force Notification System and Community Interface.
- **Phase 3 (Enhancement):** Video Gallery, Landscape-based prediction AI, and Map-based historical heatmaps

3. Specific Requirements

3.1 External Interface Requirements

The application's functionality is dependent on seamless communication with third party cloud infrastructures:

- **Firebase Authentication & Data Services:** The system interfaces with the Firebase SDK to manage encrypted user credentials and real-time data synchronization. This ensures a secure session for engineers and reliable storage for blasting logs.
- **Google Maps Platform API:** The application communicates with Google's geospatial servers to fetch high-resolution satellite imagery and topographic data, enabling the GIS-based site selection feature.

□

3.2 Functional Requirements

3.2.1 Functional Requirements

- **F01: Predictive Modelling Engine:** The system shall employ a mathematical algorithm that accepts a minimum of two input variables (e.g., **Area** and **Explosive Type**) to predict critical outputs, including **Hole Depth**, **Hole Spacing**, and **Total Quantity**. The engine must account for the selected **Landscape Profile** (Rocky vs. Mountainous) to adjust these predictions.
- **F02: Automated Data Logging:** Upon the confirmation of a calculation, the system must automatically archive the session into a non-volatile **History Tab**. This record shall include the timestamp, coordinates, landscape type, and all calculated blasting parameters.
- **F03: Community Alerting System:** The system shall provide an authorized Engineer with a "Blast Alert" trigger. Once activated, the system must broadcast a high-priority push notification to all users categorized as "Community Members" within a specific geographic radius of the blast site.
- **F0: ID Requirement Statement Priority Firebase Service**
- FR-004 The system shall allow blasting workers to register/login with email/password. High Firebase Auth
- FR-005 The system shall accept rock density, rock strength, blast area size, fracture point, water content, and elasticity as inputs. High local state.
- FR-006 The system shall calculate and display explosive type (e.g., ANFO, emulsion). High Fire store (store result)
- FR-007 The system shall calculate burden, spacing, hole depth, and number of detonators. High Fire store.
- FR-008 The system shall calculate the amount of oil/gravel for stemming. High Fire store
- FR-009 The system shall calculate a danger level (Low/Medium/High) based on explosive type and water content. High Fire store.
- FR-010 The system shall calculate estimated fragmentation throw distance in meters. High Fire store.
- FR-011 The system shall allow a mine operator to schedule a blast with time, location, and safety radius. High Fire store.
- FR-012 The system shall send a push notification to community members within the blast radius at least 1 hour before blast time. High FCM + Fire store.
- FR-013 The system shall save every blast calculation to the user's history, ordered by date. Medium Fire store

3.2.2 Predictive Calculation Logic & Formulas

The following script defines how the application processes inputs to determine blasting parameters and material requirements.

A. Environmental Multipliers (Landscape Profiles)

The calculation engine applies a **Geological Constant (\$K\$)** based on the rock density and landscape profile selected by the user:

□ **Soft Rock / Rocky Area:** $K = 0.35$ □ **Hard Rock / Mountainous Area:**
 $K = 0.50$ □ **Standard Ground:** $K = 0.40$

B. Primary Parameter Script

The system shall calculate the outputs using these standardized engineering equations:

1. Burden Calculation (\$B\$):

$$B = d \times K$$

(Where d is the hole diameter in mm and K is the Burden Factor) 2.

Spacing (\$S\$):

$$S = B \times 1.25$$

(Based on a standard equilateral blasting pattern) 3.

Hole Depth (\$H\$):

$$H = B \times 2.5$$

4. Sub-drilling (\$J\$):

$$J = B \times 0.3$$

5. Total Quantity of Explosives (\$W_e\$):

$$W_e = (B \times S \times H) \times PF$$

6. Efficiency

Power factor

$$PF = Q/V$$

Specific charge

$$SC = Q/V$$

Landscape / Ground Condition	Water Content	Recommended Explosive Type
Mountainous (Hard Rock)	Dry	ANFO
Mountainous (Hard Rock)	Wet / High Moisture	Emulsion Explosives
Rocky (Soft/Medium)	Dry	Heavy ANFO
Rocky (Soft/Medium)	Wet / High Moisture	Slurry / Water Gel

3.3 Performance Requirements

- **P01: Computational Latency:** To ensure field efficiency, the predictive engine must complete all local processing and render results on the user interface in **under 2.0 seconds**.
- **P02: Transmission Latency:** Under standard network conditions, the "Force Notification" safety alert must reach the cloud distribution layer and initiate delivery to end-user devices within **5 seconds** of the trigger.

3.4 Logical Database Requirements

The application uses **Firestore**, a NoSQL document-oriented database, structured to ensure rapid retrieval and offline persistence. The schema is organized into the following primary collections:

- **Users Collection:** Contains `User_ID`, `Role_Type` (*Engineer or Community*), `Email`, and `Account_Created_Date`.
- **BlastLogs Collection:** A spatial-enabled collection containing `GPS_Coordinates`, `Calculated_Depth`, `Spacing_Intervals`, `Landscape_ID`, and `Engineer_ID`.
- **CommunityAlerts Collection:** A log of all safety broadcasts, containing `Timestamp`, `Blast-Origin_Coordinates`, and `Notification_Status`.

3.5 Design Constraints

- **C01: Standard Compliance:** The user interface must adhere strictly to the **Material Design** standards for Android and the **Human Interface Guidelines (HIG)** for iOS. This ensures a predictable and accessible experience for users in high-stress mining environments.

3.6 Software System Attributes

- **Reliability:** To mitigate the risk of physical hazards, all mathematical formulas used within the calculator must undergo independent **peer review and validation** against established mining engineering standards before deployment.
- **Availability:** The application must maintain **100% availability** for core offline functions (calculator and local GPS logging). The system must remain operational regardless of the availability of external network services.
- **Security:** All sensitive data is protected via **AES encryption** through Firebase. A strict **Role Based Access Control (RBAC)** model is enforced, ensuring that community members are technically restricted from accessing or viewing internal engineering data.
- **Maintainability:** The software shall be developed using a **Modular Architecture**. This allows for the addition of new explosive materials or geological formulas without requiring a comprehensive rewrite of the application's core logic.
- **Portability:** The application shall be developed using a **Cross Platform Framework** (such as Flutter or React Native), ensuring a consistent codebase that is compatible with both the Android and iOS ecosystems.

3.7 Functional Hierarchy

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